

Question 1:

Two distinct numbers a and b are selected at random from $1, 2, 3, \dots, 50$. The probability that their product ab is divisible by 3 is

- (A) $\frac{8}{25}$
- (B) $\frac{561}{1225}$
- (C) $\frac{664}{1225}$
- (D) $\frac{272}{1225}$

Question 2:

If a random variable x has the probability distribution

x	0	1	2	3	4	5	6	7
$P(x)$	0	$2k$	k	$3k$	$2k^2$	$2k$	$k^2 + k$	$7k^2$

then $P(3 < x \leq 6)$ is equal to

- (A) 0.22
- (B) 0.33
- (C) 0.34
- (D) 0.64

Question 3:

Let $f: [1, \infty) \rightarrow \mathbb{R}$ be a differentiable function. If $6 \int_1^x f(t) dt = 3xf(x) + x^3 - 4$ for all $x \geq 1$, then the value of $f(2) - f(3)$ is

- (A) 3
- (B) -4
- (C) -3
- (D) 4

Question 4:

If the image of the point $P(1, 2, a)$ in the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{2}$ is $Q(5, b, c)$, then $a^2 + b^2 + c^2$ is equal to

- (A) 293
- (B) 298
- (C) 264

- (D) 283

Question 5:

Let A be a 3×3 matrix such that $A + A^T = O$. If

$$A[1-10] = [332], A_2[1-10] = [-319-24]$$

$$\text{and } \det(\text{adj}(2\text{adj}(A+I))) = 2^\alpha 3^\beta 11^\gamma,$$

where α, β, γ are non-negative integers, then the value of $\alpha + \beta + \gamma$ is _____.

Question 6:

A projectile is thrown upward at an angle 60° with the horizontal. The speed of the projectile is 20 m/s when its direction of motion is 45° with the horizontal. The initial speed of the projectile is m/s.

- (A) $20\sqrt{2}$
- (B) 40
- (C) $20\sqrt{3}$
- (D) $40\sqrt{2}$

Question 7:

Three identical coils C_1 , C_2 and C_3 are closely placed such that they share a common axis. C_2 is exactly midway. C_1 carries current I in anti-clockwise direction while C_3 carries current I in clockwise direction. An induced current flows through C_2 will be in clockwise direction when

- (A) C_1 and C_3 move with equal speeds away from C_2
- (B) C_1 moves away from C_2 and C_3 moves towards C_2
- (C) C_1 moves towards C_2 and C_3 moves away from C_2
- (D) C_1 and C_3 move with equal speeds towards C_2

Question 8:

Given below are two statements:

Statement I: Pressure of a fluid is exerted only on a solid surface in contact as the fluid-

pressure does not exist everywhere in a still fluid.

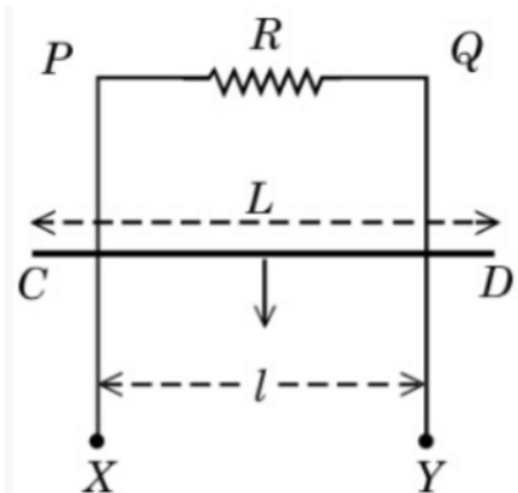
Statement II: Excess potential energy of the molecules on the surface of a liquid, when compared to interior, results in surface tension.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are false
- (B) Statement I is true but Statement II is false
- (C) Both Statement I and Statement II are true
- (D) Statement I is false but Statement II is true

Question 9:

XPQY is a vertical smooth long loop having a total resistance R , where PX is parallel to QY and the separation between them is l . A constant magnetic field B perpendicular to the plane of the loop exists in the entire space. A rod CD of length L ($L > l$) and mass m is made to slide down from rest under gravity as shown. The terminal speed acquired by the rod is _____ m/s.



- (A) mgR/B^2l^2
- (B) $2mgR/B^2L^2$
- (C) $8mgR/B^2l^2$

- (D) $2mgR/B2I2$

Question 10:

A circular disc has radius R_1 and thickness T_1 . Another circular disc made of the same material has radius R_2 and thickness T_2 . If the moments of inertia of both the discs are same and $R_1 R_2 = 2$, then $T_1 T_2 = 1\alpha$.

The value of α is _____.

Question 11:

As compared with chlorocyclohexane, which of the following statements correctly apply to chlorobenzene?

[label=\Alph.]*

The magnitude of negative charge is more on chlorine atom.

The C--Cl bond has partial double bond character.

C--Cl bond is less polar.

C--Cl bond is longer due to repulsion between delocalised electrons of the aromatic ring and lone pairs of electrons of chlorine.

The C--Cl bond is formed using sp^2 hybridised orbital of carbon.

Choose the correct answer from the options given below:

- (A) B, C and D Only
- (B) A, C and E Only
- (C) A, D and E Only
- (D) B, C and E Only

Question 12:

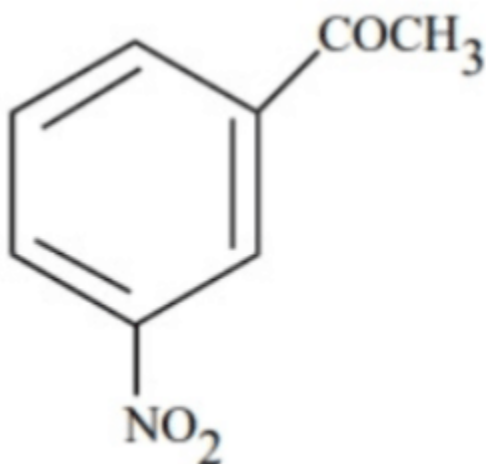
Consider the transition metal ions Mn^{3+} , Cr^{3+} , Fe^{3+} and Co^{3+} and all form low spin octahedral complexes. The correct decreasing order of unpaired electrons in their respective d-orbitals of the complexes is

- (A) $Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$
- (B) $Fe^{3+} > Co^{3+} > Mn^{3+} > Cr^{3+}$
- (C) $Mn^{3+} > Fe^{3+} > Co^{3+} > Cr^{3+}$
- (D) $Cr^{3+} > Fe^{3+} > Co^{3+} > Mn^{3+}$

Question 13:

Given below are two statements:

Statement I: Benzene is nitrated to give nitrobenzene, which on further treatment with $\text{CH}_3\text{COCl}/\text{AlCl}_3$ will give the product shown.



Statement II: $-\text{NO}_2$ group is a meta-directing and deactivating group.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Statement I is correct but Statement II is incorrect
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is incorrect but Statement II is correct
- (D) Both Statement I and Statement II are correct

Question 14:

Given below are two statements:

Statement I: The Henry's law constant K_H is constant with respect to variations in solution concentration over the range for which the solution is ideally dilute.

Statement II: K_H does not differ for the same solute in different solvents.

In the light of the above statements, choose the correct answer from the options given below.

- *(A) Both Statement I and Statement II are false*
- *(B) Statement I is false but Statement II is true*
- *(C) Both Statement I and Statement II are true*
- *(D) Statement I is true but Statement II is false*

Question 15:

Sodium fusion extract of an organic compound (Y) with CHCl_3 and chlorine water gives violet colour to the CHCl_3 layer. 0.15g of (Y) gave 0.12g of the silver halide precipitate in Carius method. Percentage of halogen in the compound (Y) is _____ (Nearest integer).

Given:

$\text{C}=12, \text{H}=1, \text{Cl}=35.5, \text{Br}=80, \text{I}=127$